

01: Introduction to Networks

Networks I: Concepts and Algorithms

Michael Fehl

2026-03-20

Table of contents

1	Why Networks?	2
2	Graph Theory: The Mathematical Language of Networks	3
2.1	From Bridges to Graphs	3
2.2	Formal Definitions	3
2.2.1	Variants Worth Knowing	4
3	Degree: The Simplest Node Summary	6
3.1	Definition and Basic Identities	6
3.2	Directed Degree	7
3.3	The Degree Distribution	7
4	Paths, Distance, and the Small World	9
4.1	Walks, Paths, and Geodesics	9
4.2	Why Distance Matters: The Small World Phenomenon	10
4.3	Computing Distances: BFS and the Facebook Networks	11
5	Centrality: Who Matters, and Why?	13
5.1	Degree Centrality Is Not Enough	13
5.2	Betweenness Centrality	13
5.2.1	The Medici Example	14
5.3	Closeness Centrality	14
5.4	Eigenvector Centrality and PageRank	14
6	Clustering: Triadic Closure and Community Structure	17
6.1	Motivation: Triangles Are Everywhere	17
6.2	Global Clustering (Transitivity)	18
6.3	Local Clustering Coefficient	18

7	The Strength of Weak Ties	20
7.1	Granovetter's Paradox	20
7.2	Triadic Closure and Local Bridges	20
7.3	LinkedIn RCT Validation	21
8	Networks in Economics: Price of Anarchy	23
8.1	Braess's Paradox	23
9	Summary and What's Next	24
9.1	What We Can Now Measure	24
9.2	The Critical Gap: Where Do These Numbers Come From?	24
9.3	References	26

1 Why Networks?

Before diving into definitions, it is worth asking: *why is the study of networks a distinct field at all?* Most economic models treat agents as interacting through aggregate prices; a firm responds to market wages, not to the specific workers it employs. Most statistical models treat observations as i.i.d. draws. Neither framework is adequate when the *identity* of who you are connected to fundamentally changes outcomes.

Consider a few canonical examples:

Setting	Nodes	Edges	Key question
Labour market	Workers, firms	Employment relations	Why do referred workers earn more?
Epidemiology	People	Physical contact	Who to vaccinate first?
Finance	Banks	Interbank loans	Which bank's failure is systemic?
Supply chain	Firms	Input-output flows	Where does a shock propagate?
Social media	Users	Follows/retweets	How does misinformation spread?

What all of these have in common: the *structure* of connections; not just their average density; determines outcomes. A disease spreading through a hub-and-spoke airline network behaves very differently from one spreading through a lattice. A worker connected to a structurally diverse set of acquaintances finds jobs faster than one embedded in a tightly clustered clique, even holding the *number* of connections fixed.

i Course perspective

This is not primarily a machine learning course. We study networks from an economics angle: we want to model strategic behaviour, predict equilibria, and use the structure of networks to understand data on economic and social activity. Machine learning tools (GNNs, embeddings) appear later as *instruments* for prediction, not as the main object of study.

2 Graph Theory: The Mathematical Language of Networks

2.1 From Bridges to Graphs

The history of graph theory begins with a practical puzzle. In 1736, the city of Königsberg (now Kaliningrad) was divided by the Pregel River into four landmasses connected by seven bridges. Citizens wondered: is there a walk that crosses each bridge *exactly once*?

Leonhard Euler proved the answer is no, and in doing so laid the foundation for graph theory. His key insight: **the problem is purely topological**. The specific geography is irrelevant; what matters is only which landmasses are connected, and how many times. He observed that such a walk (now called an *Eulerian path*) requires every node except the start and end to have an even number of edges. Königsberg has four nodes, all with odd degree; so no such walk exists.

The abstraction Euler introduced; a set of *nodes* connected by *edges*; is exactly the graph. Every network model in this course is a specialisation of this same idea.

2.2 Formal Definitions

A **graph** $G = (V, E)$ consists of:

- V : a set of **vertices** (nodes), $|V| = n$
- E : a set of **edges** (links), $|E| = m$

For an **undirected, unweighted** graph, edges are unordered pairs $\{i, j\} \in E$, and the adjacency matrix is:

$$A_{ij} = \begin{cases} 1 & \text{if } \{i, j\} \in E \\ 0 & \text{otherwise} \end{cases}$$

Three key properties of this matrix:

1. **Binary:** $A_{ij} \in \{0, 1\}$
2. **Symmetric:** $A_{ij} = A_{ji}$ (since edges have no direction)
3. **Zero diagonal:** $A_{ii} = 0$ (no self-loops)

2.2.1 Variants Worth Knowing

Directed graphs: edges are ordered pairs $(i, j) \in E$, so A is no longer symmetric. Useful for citations, Twitter follows, river flows.

Weighted graphs: $A_{ij} = w_{ij} \in \mathbb{R}$ records edge strength (trade volume, message frequency, road capacity).

Bipartite graphs: V is partitioned into two disjoint sets V_1, V_2 with edges only between sets (actors films, buyers sellers, genes diseases).

Multigraphs: multiple edges between the same pair of nodes are allowed. We will mostly stay away from these, but they appear naturally in citation data.

```
par(mfrow = c(1, 3), mar = c(1, 1, 3, 1))

## ---- Undirected ----
g_un <- graph(
  edges = c(0,1, 1,2, 2,3, 3,4, 4,5, 0,5, 1,4) + 1L, # 1-indexed
  n = 6, directed = FALSE
)
V(g_un)$label <- as.character(0:5)
plot(
  g_un,
  layout      = layout_in_circle(g_un),
  vertex.size = 20,
  vertex.color = "#4C72B0",
  vertex.label = V(g_un)$label,
  vertex.label.color = "white",
  vertex.label.cex = 0.9,
  edge.color   = "#666666",
  main        = "Undirected, Unweighted"
)

## ---- Directed ----
g_di <- graph(
  edges = c(0,1, 1,2, 2,3, 3,4, 4,5, 5,0, 1,4) + 1L,
  n = 6, directed = TRUE
)
```

```

V(g_di)$label <- as.character(0:5)
plot(
  g_di,
  layout      = layout_in_circle(g_di),
  vertex.size = 20,
  vertex.color = "#DD8452",
  vertex.label = V(g_di)$label,
  vertex.label.color = "white",
  vertex.label.cex = 0.9,
  edge.color    = "#666666",
  edge.arrow.size = 0.5,
  main          = "Directed"
)

## ---- Bipartite ----
g_bi <- make_full_bipartite_graph(3, 3)
V(g_bi)$label <- c("A", "B", "C", "1", "2", "3")
V(g_bi)$color <- c(rep("#55A868", 3), rep("#C44E52", 3))
# Simple bipartite layout
lay_bi <- layout_as_bipartite(g_bi)
plot(
  g_bi,
  layout      = lay_bi,
  vertex.size = 20,
  vertex.color = V(g_bi)$color,
  vertex.label = V(g_bi)$label,
  vertex.label.color = "white",
  vertex.label.cex = 0.9,
  edge.color    = "#666666",
  main          = "Bipartite"
)

```

Undirected, Unweighted

Directed

Bipartite

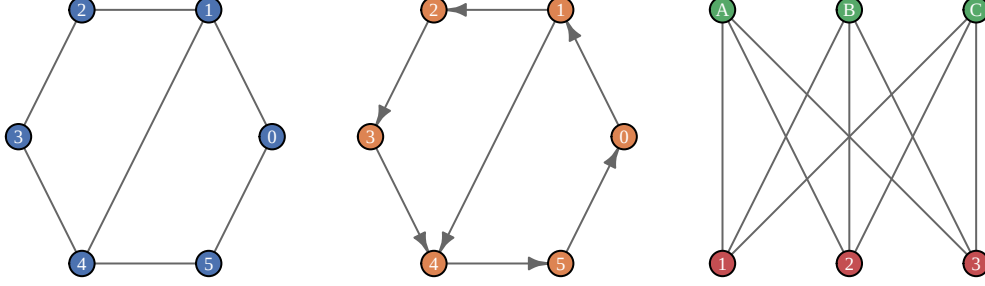


Figure 1: Three graph variants on 6 nodes

3 Degree: The Simplest Node Summary

3.1 Definition and Basic Identities

Describing a network by its full adjacency matrix A is informative but unwieldy. The first summary statistic is the **degree** of a node: the number of edges it has.

$$k_i = \sum_{j=1}^n A_{ij}$$

For weighted networks this generalises to **strength**: $s_i = \sum_j A_{ij} w_{ij}$.

Two fundamental identities follow immediately from the definition:

Handshaking lemma: every edge contributes one to each endpoint's degree, so the sum of degrees counts each edge twice:

$$\sum_{i=1}^n k_i = 2m \implies \langle k \rangle = \frac{2m}{n}$$

Density: the fraction of possible edges that actually exist:

$$\rho = \frac{m}{\binom{n}{2}} = \frac{2m}{n(n-1)} = \frac{\langle k \rangle}{n-1}$$

Real networks are almost always *sparse*: $\rho \rightarrow 0$ as $n \rightarrow \infty$, meaning $m = O(n)$ rather than $m = O(n^2)$. This has enormous computational implications — it is what makes sparse matrix formats and igraph’s C-level operations practical.

3.2 Directed Degree

In a directed graph, a node has three degree notions:

$$k_i^{\text{out}} = \sum_j A_{ij}, \quad k_i^{\text{in}} = \sum_j A_{ji}, \quad k_i^{\text{tot}} = |\{j : A_{ij} = 1 \text{ or } A_{ji} = 1\}|$$

The conservation law: $\sum_i k_i^{\text{out}} = \sum_i k_i^{\text{in}} = m$, so average in-degree always equals average out-degree.

In citation networks (PS1 Q4): k_i^{out} = number of papers cited by i (bibliography size); k_i^{in} = number of papers that cite i (citations received, often called “in-degree” or “citation count”).

3.3 The Degree Distribution

A single number like $\langle k \rangle$ conceals crucial heterogeneity. The **degree distribution** $p(k) = \Pr(K = k)$ gives the full picture.

Why does this matter? Consider two networks, both with $n = 1490$ nodes and $\langle k \rangle = 25.6$ (the 2004 political blogs network studied by Adamic & Glance). The maximum degree in that network was $k_{\text{max}} = 351$; more than 13 times the mean. Linear-scale histograms make this invisible. The right tool is a **log–log plot** of the complementary CDF:

$$\bar{F}(k) = \Pr(K \geq k)$$

If this is linear on a log–log scale, the degree distribution follows a **power law**: $p(k) \propto k^{-\gamma}$. We will study why this emerges and what it implies in Lecture 2 (Section 4: Preferential Attachment).

```
## Load American75 .mat file
mat <- readMat("data/facebook_small/American75.mat")
A_sp <- mat$A # sparse matrix (dgCMatrix or similar)

# Build igraph from sparse adjacency
A_trip <- as(A_sp, "TsparseMatrix") # converts to triplet form (i, j, x)
edges_fb <- cbind(A_trip@i + 1L, A_trip@j + 1L) # 1-indexed
g <- graph_from_edgelist(edges_fb, directed = FALSE)
```

```

g <- simplify(g)                                # remove duplicates / self-loops

degs    <- degree(g)
n_g     <- vcount(g)
mean_k  <- mean(degs)

## CCDF
ccdf_k  <- sort(unique(degs))
ccdf_val <- sapply(ccdf_k, function(k) mean(degs >= k))

## Plots
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))

# Left: linear histogram
hist(
  degs, breaks = 40,
  col    = "#4C72B0", border = "white",
  xlab   = "Degree k",
  ylab   = "Count",
  main   = sprintf("Degree histogram\n(n=%d, <k>=%.1f)", n_g, mean_k)
)

# Right: log-log CCDF
plot(
  ccdf_k, ccdf_val,
  log    = "xy",
  pch    = 19, cex = 0.5, col = adjustcolor("#C44E52", alpha.f = 0.7),
  xlab   = "Degree k",
  ylab   = expression(bar(F)(k) == Pr(K >= k)),
  main   = "Complementary CDF (log-log)"
)
grid()

```

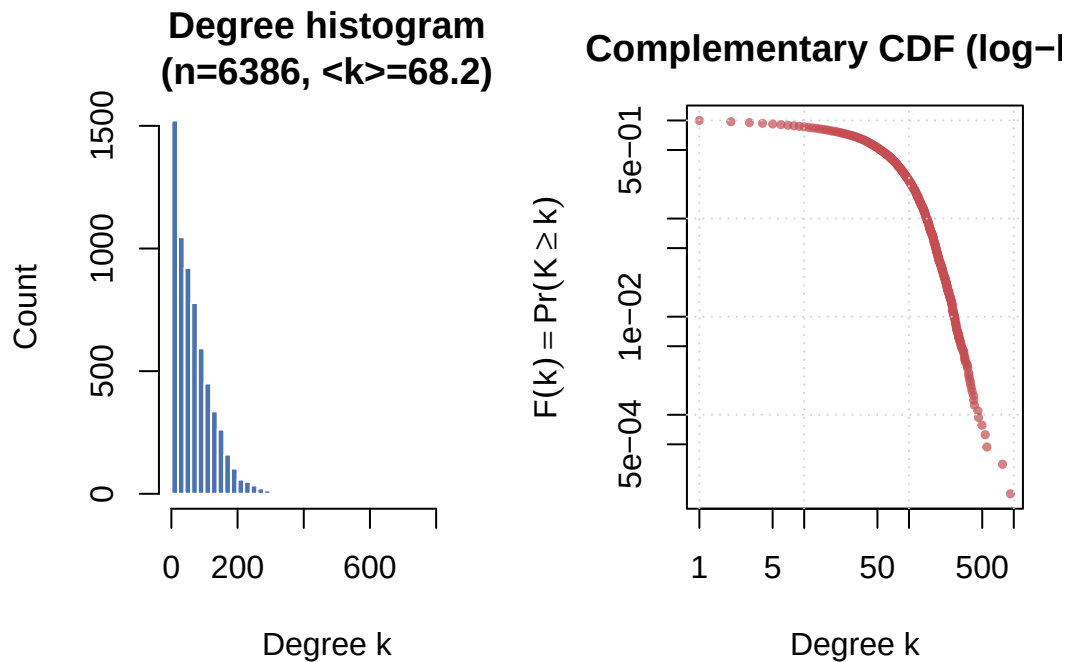



Figure 2: Degree distribution of the American75 Facebook network (log–log scale).

```
cat(sprintf("n = %d, m = %d\n", vcount(g), ecount(g)))
```

```
n = 6386, m = 217662
```

```
cat(sprintf("<k> = %.2f, k_max = %d, k_min = %d\n",
            mean(degs), max(degs), min(degs)))
```

```
<k> = 68.17, k_max = 930, k_min = 1
```

```
cat(sprintf("Density = %.5f\n", graph.density(g)))
```

```
Density = 0.01068
```

4 Paths, Distance, and the Small World

4.1 Walks, Paths, and Geodesics

Moving from *who* is connected (degree) to *how* things flow requires thinking about paths through the network.

Formally, for an undirected graph:

- A **walk** is any sequence of adjacent nodes $i_1, i_2, \dots, i_k$ where $\{i_t, i_{t+1}\} \in E$ for all t . Nodes and edges may repeat.
- A **path** is a walk where all nodes are distinct.
- A **cycle** is a closed path: $i_1 = i_k$ with all intermediate nodes distinct.
- The **geodesic distance** ℓ_{ij} is the length of the *shortest path* between i and j . If no path exists, $\ell_{ij} = \infty$.

The **diameter** of a graph is the longest geodesic across all pairs:

$$\text{diam}(G) = \max_{i \neq j} \ell_{ij}$$

where we exclude $\ell_{ij} = \infty$ (i.e., we measure within the largest connected component).

The **mean geodesic distance** (average path length) is:

$$\langle \ell \rangle = \frac{1}{n(n-1)} \sum_{i \neq j} \ell_{ij}$$

4.2 Why Distance Matters: The Small World Phenomenon

In 1967, sociologist Stanley Milgram asked 296 residents of Wichita and Omaha to route a letter to a specific target in Boston; but only by passing it to someone they knew personally. Of the 64 letters that arrived, the median chain length was **5.5 intermediaries**; giving rise to the phrase “six degrees of separation.”

(The study has since been critiqued: the 78% non-completion rate is not random, and the participants who succeeded were likely unusually well-connected. But the qualitative finding holds in larger digital datasets.)

A back-of-envelope argument explains why short paths are possible. Suppose each person has γ friends, with no overlap between social circles (a strong assumption). Then:

- At distance 1: you can reach γ people
- At distance 2: γ^2 people
- At distance d : γ^d people

Setting $\gamma^d = n$ and solving: $d = \frac{\ln n}{\ln \gamma}$. For the US population $n \approx 3 \times 10^8$:

$$d = \frac{\ln(3 \times 10^8)}{\ln \gamma}$$

Milgram's result of $d \approx 6$ implies $\gamma \approx \exp\left(\frac{\ln(3 \times 10^8)}{6}\right) \approx 53$ friends. That seems large; but notice Facebook mutual friends don't count (weak ties, not acquaintances). The calculation is sensitive to the definition of "friend."

! The key takeaway

Networks are "small" when the diameter $D = O(\log n)$ rather than $O(n)$. This exponential expansion along paths is what makes diseases, information, and prices propagate so quickly through real-world networks.

4.3 Computing Distances: BFS and the Facebook Networks

Breadth-first search (BFS) from a single source computes all distances in $O(n + m)$ time. For large graphs, computing all $\binom{n}{2}$ pairwise distances by running BFS from every node costs $O(n(n + m))$; feasible for $n \lesssim 10^4$, but not for the full Facebook100 dataset.

PS1, Question 1 asks you to compute diameter and mean geodesic distance for each of the 84 Facebook university networks. The key approximation: *sample* s random source nodes and average; the error is $O(1/\sqrt{s})$.

```
## ---- Helper: sample average path length ----
sample_apl <- function(g, n_samples = 200, seed = 42) {
  set.seed(seed)
  comps <- components(g)
  lcc <- induced_subgraph(g, which(comps$membership == which.max(comps$csizes)))

  n_lcc <- vcount(lcc)
  sources <- sample(seq_len(n_lcc), min(n_samples, n_lcc))

  total <- 0; count <- 0
  for (src in sources) {
    dists <- distances(lcc, v = src, mode = "out")
    finite_d <- dists[dists > 0 & is.finite(dists)]
    total <- total + sum(finite_d)
    count <- count + length(finite_d)
  }
  if (count == 0) return(NA_real_)
  total / count
}

## ---- Helper: double-sweep diameter lower bound ----
approx_diameter <- function(g) {
```

```

comps <- components(g)
lcc <- induced_subgraph(g, which(comps$membership == which.max(comps$csize)))

# Step 1: BFS from node 1
d1 <- distances(lcc, v = 1)
v <- which.max(d1)

# Step 2: BFS from v
d2 <- distances(lcc, v = v)
max(d2[is.finite(d2)])
}

t0 <- proc.time()["elapsed"]
apl <- sample_apl(g, n_samples = 300)
diam <- approx_diameter(g)
t1 <- proc.time()["elapsed"]

comps <- components(g)
lcc <- induced_subgraph(g, which(comps$membership == which.max(comps$csize)))

cat(sprintf("Network: American75\n"))

```

Network: American75

```
cat(sprintf("  Nodes (LCC): %d\n", vcount(lcc)))
```

Nodes (LCC): 6370

```
cat(sprintf("  Edges (LCC): %d\n", ecount(lcc)))
```

Edges (LCC): 217654

```
cat(sprintf("  <l> ~ %.3f (sampled)\n", apl))
```

<l> ~ 2.795 (sampled)

```
cat(sprintf("  diam ~ %d (double-sweep lower bound)\n", diam))
```

diam ~ 9 (double-sweep lower bound)

```
cat(sprintf("  Computed in %.2fs\n", t1 - t0))
```

Computed in 3.39s

```
cat(sprintf("\n  log(n)/log(<k>) = %.2f  (ER prediction)\n",  
          log(vcount(lcc)) / log(mean(degree(lcc)))))
```

log(n)/log(<k>) = 2.07 (ER prediction)

5 Centrality: Who Matters, and Why?

5.1 Degree Centrality Is Not Enough

Degree captures *local* connectivity; how many direct neighbours a node has. But in many settings, influence is determined by *position* in the network structure, not just by raw connectivity.

Consider a node with degree 5 sitting at the junction between two otherwise disconnected communities, versus a node with degree 10 embedded in a tightly knit clique. The former is far more influential for information flow, despite lower degree.

This motivates centrality measures that capture *global* structural importance.

5.2 Betweenness Centrality

The **betweenness centrality** of node k counts how often k lies on the shortest path between other pairs of nodes:

$$B_k = \frac{1}{\binom{n-1}{2}} \sum_{\substack{i \neq j \\ k \notin \{i, j\}}} \frac{P_k(i, j)}{P(i, j)}$$

where $P(i, j)$ is the number of geodesic paths between i and j , and $P_k(i, j)$ is the number that pass through k . The normalisation by $\binom{n-1}{2}$ makes $B_k \in [0, 1]$.

Interpretation: a node with high betweenness is a *broker*; it controls traffic between otherwise separate parts of the network. Removing such nodes tends to dramatically fragment the graph.

5.2.1 The Medici Example

15th-century Florence witnessed the rise of the Medici family from relative obscurity to political and economic dominance. The puzzle: many other families were wealthier and had longer aristocratic pedigrees.

Padgett & Ansell (1993) digitised the marriage network of Florentine families and computed betweenness centralities. The Medici had $B_{\text{Medici}} = 0.522$ — the highest of any family, by a wide margin. The Medici were positioned at the structural centre of the marriage network, serving as the broker connecting families that could not directly influence each other. Their political rise was not incidental: it was written into the topology of the network.

💡 Betweenness as an economic asset

High betweenness creates *information brokerage*: brokers hear news, prices, and opportunities before they diffuse to the rest of the network. It also creates *structural holes* (Burt 1992): the broker can play parties off against each other when they cannot coordinate directly.

5.3 Closeness Centrality

Closeness centrality measures how quickly a node can reach all others:

$$C_k = \frac{n-1}{\sum_{j \neq k} \ell_{kj}}$$

A node with high closeness sits near the “centre of gravity” of the network and can spread information to all others quickly. Unlike betweenness, it does not depend on being a bottleneck; it rewards being well-connected on average.

5.4 Eigenvector Centrality and PageRank

Both betweenness and closeness treat all paths equally. But a link from a high-status node should count more than a link from a peripheral one. This motivates **eigenvector centrality**:

$$x_i = \frac{1}{\lambda} \sum_j A_{ij} x_j \iff Ax = \lambda x$$

The centrality vector x is the leading eigenvector of A (by the Perron–Frobenius theorem, this has all non-negative entries for connected graphs). Each node’s score is proportional to the sum of its neighbours’ scores.

PageRank (Brin & Page, 1998) adds a damping factor to handle dangling nodes and make the walk ergodic:

$$r_i = \frac{1-d}{n} + d \sum_j A_{ji} \frac{r_j}{k_j^{\text{out}}}$$

With damping $d = 0.85$: with probability d a random walker follows an outgoing link; with probability $1 - d$ it teleports to a random node. PageRank is the stationary distribution of this random walk.

```
family_names <- c(
  "Acciaiuoli", "Albizzi", "Barbadori", "Bischeri", "Castellani",
  "Ginori", "Guadagni", "Lamberteschi", "Medici", "Pazzi",
  "Peruzzi", "Pucci", "Ridolfi", "Salviati", "Strozzi",
  "Tornabuoni"
)

# Edges from Padgett (1993); converted to 1-indexed
edges_flo <- matrix(c(
  1,9, 2,7, 2,9, 2,10, 3,9, 3,5, 4,7, 4,12, 4,15,
  5,9, 5,15, 6,9, 7,11, 7,14, 7,15, 8,7, 9,10, 9,13,
  9,14, 9,16, 11,15, 13,15, 13,16, 15,16
), ncol = 2, byrow = TRUE)

g_flo <- graph_from_edgelist(edges_flo, directed = FALSE)
g_flo <- set_vertex_attr(g_flo, "name", value = family_names)
medici_idx <- which(family_names == "Medici") # = 9

# Compute centralities
deg_c <- degree(g_flo)
btw_c <- betweenness(g_flo, normalized = TRUE)
cls_c <- closeness(g_flo, normalized = TRUE)
pr_c <- page_rank(g_flo, damping = 0.85)$vector

## Plot; 2x2 horizontal bar charts
old_par <- par(mfrow = c(2, 2), mar = c(3, 7, 3, 1))

plot centrality <- function(vals, title, color) {
  bar_colors <- rep(color, length(family_names))
```

```

bar_colors[medici_idx] <- "#2d2d2d"
bp <- barplot(
  vals, names.arg = family_names, horiz = TRUE,
  col = bar_colors, border = NA,
  las = 2, cex.names = 0.7,
  main = paste0(title, " Centrality\n(Medici in black)"),
  xlab = ""
)
abline(v = vals[medici_idx], lty = 2, col = "#2d2d2d", lwd = 1.5)
}

plot_centrality(deg_c, "Degree", "#4C72B0")
plot_centrality(btw_c, "Betweenness", "#C44E52")
plot_centrality(cls_c, "Closeness", "#55A868")
plot_centrality(pr_c, "PageRank", "#DD8452")

```

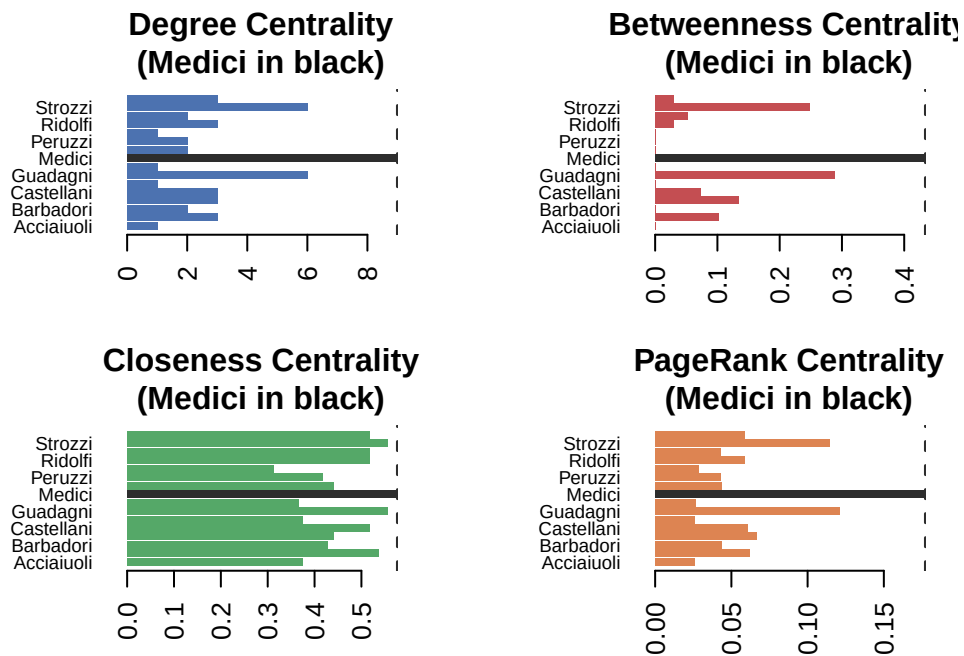


Figure 3: Centrality measures on the Florentine marriage network. The Medici (node 8, 1-indexed) dominate by betweenness.

```

par(old_par)

cat("Medici centralities:\n")

```


Medici centralities:

```
cat(sprintf(" Degree:      %d\n",      deg_c[medici_idx]))
```

Degree: 9

```
cat(sprintf(" Betweenness: %.4f (highest in network: %.4f)\n",  
            btw_c[medici_idx], max(btw_c)))
```

Betweenness: 0.4341 (highest in network: 0.4341)

```
cat(sprintf(" Closeness:   %.4f\n", cls_c[medici_idx]))
```

Closeness: 0.5769

```
cat(sprintf(" PageRank:    %.4f\n", pr_c[medici_idx]))
```

PageRank: 0.1773

6 Clustering: Triadic Closure and Community Structure

6.1 Motivation: Triangles Are Everywhere

Degree and betweenness describe *individual* nodes. A different question asks about the *local structure* around a node: to what extent are my neighbours also connected to each other?

This matters because:

1. **Trust propagation:** if A trusts B and A trusts C, there is social pressure for B and C to also form a relationship; what Simmel called the principle of *triadic closure*.
2. **Community detection:** clusters in networks are regions of high local clustering separated by sparse cross-community edges.
3. **Model validation:** the Erdős–Rényi model predicts $C \rightarrow 0$ as $n \rightarrow \infty$, but real networks maintain high clustering. This falsifies ER as a realistic model.

6.2 Global Clustering (Transitivity)

The **global clustering coefficient** (or transitivity) is the fraction of connected triples that are closed into triangles:

$$C(A) = \frac{\text{number of triangles} \times 3}{\text{number of connected triples}} = \frac{\sum_{ijk} A_{ij} A_{jk} A_{ki}}{\sum_{ijk} A_{ij} A_{jk}}$$

The factor of 3 accounts for the three different “pivot” nodes in each triangle (each triangle contributes to three distinct connected triples). This makes $C \in [0, 1]$.

In matrix terms, the numerator is $\text{tr}(A^3)$ (each walk $i \rightarrow j \rightarrow k \rightarrow i$ of length 3 is a triangle), and the denominator is $\sum_i k_i(k_i - 1)$.

6.3 Local Clustering Coefficient

The **local clustering coefficient** for node i with degree $k_i \geq 2$ is:

$$C_i(A) = \frac{\text{edges among neighbours of } i}{\binom{k_i}{2}} = \frac{2L_i}{k_i(k_i - 1)}$$

where L_i is the number of edges among i 's k_i neighbours.

The **average clustering coefficient** $\bar{C} = \frac{1}{n} \sum_i C_i$ weights all nodes equally regardless of degree, giving more weight to low-degree nodes compared to the global $C(A)$.

i C vs $\bar{C}$: which to use?

$C(A)$ (global transitivity) is the right metric when you care about the fraction of triangles in the graph as a whole. $\bar{C}$ is more sensitive to the local structure around typical low-degree nodes. They can differ substantially: hubs with high degree contribute little to $\bar{C}$ (since their many neighbours are unlikely to all be connected) but dominate $C(A)$.

```
clustering_report <- function(g, name) {  
  c_global <- transitivity(g, type = "global")  
  c_avg    <- transitivity(g, type = "average")    # avg local CC, isolates excluded  
  deg      <- degree(g)  
  cc_local <- transitivity(g, type = "local")  
  
  valid <- which(deg >= 2 & !is.nan(cc_local))  
  corr  <- cor(deg[valid], cc_local[valid])  
}
```

```

cat(sprintf("%s:\n", name))
cat(sprintf("  Global transitivity C(A) = %.4f\n", c_global))
cat(sprintf("  Avg local CC  C_bar      = %.4f\n", c_avg))
cat(sprintf("  Corr(degree, local CC)   = %.4f%s\n",
            corr,
            ifelse(corr < 0, " (hubs less clustered)", "")))
}

clustering_report(g_flo, "Florentine marriages")

```

Florentine marriages:

```

Global transitivity C(A) = 0.2118
Avg local CC  C_bar      = 0.4792
Corr(degree, local CC)   = -0.5938 (hubs less clustered)

```

```
clustering_report(g, "American75 (Facebook)")
```

American75 (Facebook):

```

Global transitivity C(A) = 0.1635
Avg local CC  C_bar      = 0.2474
Corr(degree, local CC)   = -0.3599 (hubs less clustered)

```

```

# Compare to ER null
c_er <- graph.density(g)
cat(sprintf("\nER null model for American75:\n"))

```

ER null model for American75:

```
cat(sprintf("  C_ER = p = %.5f (theoretically C -> 0 as n -> inf)\n", c_er))
```

```
C_ER = p = 0.01068 (theoretically C -> 0 as n -> inf)
```

```
cat(sprintf("  Actual C = %.4f\n", transitivity(g, type = "global")))
```

```
Actual C = 0.1635
```

```
cat(sprintf("  Ratio actual/ER = %.1fx  <- much higher than random!\n",
           transitivity(g, type = "global") / c_er))
```

```
Ratio actual/ER = 15.3x  <- much higher than random!
```

7 The Strength of Weak Ties

7.1 Granovetter's Paradox

One of the most celebrated findings in sociology (Granovetter 1973, the most cited paper in the sociology of networks) concerns an apparent paradox in job referrals:

Workers in the labour market find jobs primarily through referrals from acquaintances; not from close friends.

Why would weaker connections be more useful? The answer lies in the topology of clustering.

7.2 Triadic Closure and Local Bridges

Define a **strong tie** as a close friendship and a **weak tie** as an acquaintance. The **Strong Triadic Closure (STC) property** says: if node A has strong ties to both B and C , then there should be *at least a weak tie* between B and C .

Intuitively: if A is close friends with both B and C , it is socially costly and informationally inconsistent for B and C to not even know each other.

A **local bridge** between A and B is an edge whose removal would increase ℓ_{AB} substantially (say, to > 2). Local bridges connect otherwise distant parts of the network.

Claim: if STC holds and A has at least two strong ties, then any local bridge involving A must be a *weak* tie.

Proof sketch: suppose $A-B$ is a strong tie and a local bridge. By STC, for every other strong neighbour C of A , there must be an edge $B-C$. But then removing $A-B$ would not increase ℓ_{BC} much (path goes through $A-C$... to B), contradicting “local bridge.” Contradiction. $\square$

Implication for job referrals: close friends move in the same social circles. They know the same people, the same managers, the same opportunities. An acquaintance (weak tie) is embedded in a *different* cluster and therefore carries *non-redundant* information about job opportunities you haven't heard of.

7.3 LinkedIn RCT Validation

Rajkumar et al. (2022) ran a massive randomised controlled trial: over 2015–2019, LinkedIn adjusted its “People You May Know” algorithm for 20+ million users. Some users received more weak-tie suggestions, others more strong-tie suggestions.

Key results: - New connections formed from weak-tie suggestions led to *more job transmissions* than those from strong-tie suggestions - But the relationship is an **inverted U**: moderately weak ties (around 10 mutual connections) were optimal - Ties with 0 mutual connections (pure strangers) were not useful either - The effect was strongest in high-technology sectors

This is a rare case where a classic sociological theory was validated at scale using a pre-registered experiment with billions of data points.

```
# Two clusters + one bridge
edges_wt <- matrix(c(
  # Cluster A (nodes 1-5): strong ties
  1,2, 2,3, 3,4, 4,5, 5,1, 1,3,
  # Cluster B (nodes 6-10): strong ties
  6,7, 7,8, 8,9, 9,10, 10,6, 6,8,
  # Local bridge: node 3 (A) -- node 7 (B)
  3,7
), ncol = 2, byrow = TRUE)

g_wt <- graph_from_edgelist(edges_wt, directed = FALSE)
V(g_wt)$label <- as.character(0:9) # match Python 0-indexed labels

# Colors
V(g_wt)$color <- c(rep("#4C72B0", 5), rep("#C44E52", 5))

# Edge colors: bridge (3-7 in 1-indexed) = green, rest = grey
bridge_eid <- get.edge.ids(g_wt, c(3, 7))
E(g_wt)$color <- "#888888"
E(g_wt)$color[bridge_eid] <- "#2CA02C"
E(g_wt)$width <- 2
E(g_wt)$width[bridge_eid] <- 3

set.seed(42)
lay_wt <- layout_with_kk(g_wt)

plot(
  g_wt,
  layout = lay_wt,
```

```

vertex.size      = 22,
vertex.color     = V(g_wt)$color,
vertex.label     = V(g_wt)$label,
vertex.label.color = "white",
vertex.label.cex = 0.85,
edge.color       = E(g_wt)$color,
edge.width       = E(g_wt)$width,
main = "Local bridge (green) must be a weak tie\nBlue = Cluster A, Red = Cluster B"
)

```

Local bridge (green) must be a weak tie
Blue = Cluster A, Red = Cluster B

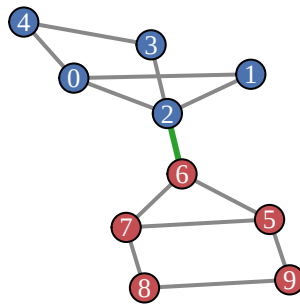


Figure 4: Illustration: local bridges between clusters must be weak ties (STC)

```

# Distance with and without the bridge
d_before <- distances(g_wt, v = 3, to = 7) # 1-indexed node 3 = Python node 2
g_wt2    <- delete_edges(g_wt, bridge_eid)
d_after  <- distances(g_wt2, v = 3, to = 7)

cat(sprintf("Distance 3->7 with bridge:    %f\n",    d_before))

```

Distance 3->7 with bridge: 1.000000

```

cat(sprintf("Distance 3->7 without bridge: %f <- bridge is critical\n", d_after))

```

Distance 3->7 without bridge: Inf <- bridge is critical

8 Networks in Economics: Price of Anarchy

The final section of the lecture places networks in an economic context. Networks are not passive structures; agents make strategic decisions *on* them. This introduces game-theoretic considerations.

8.1 Braess's Paradox

Consider a road network where one unit of traffic must travel from s to t . Two routes exist:

- Route 1: $s \xrightarrow{x_1} A \xrightarrow{1} t$, cost = $x_1 + 1$
- Route 2: $s \xrightarrow{1} B \xrightarrow{x_2} t$, cost = $1 + x_2$

where x_1, x_2 are flows on the congested links (cost = flow on those links).

At the **social optimum**: split traffic equally. $x_1 = x_2 = 1/2$. Total cost = $\frac{3}{2}$.

At the **Nash equilibrium**: each infinitesimal driver chooses the route with lowest cost *given others' choices*. Equilibrium also has $x_1 = x_2 = 1/2$ and total cost = $\frac{3}{2}$.

Now add a zero-cost road from A to B . Three routes now exist, including $s \rightarrow A \rightarrow B \rightarrow t$ with cost $x_1 + 0 + x_2$. At the *new* Nash equilibrium: everyone takes the $s \rightarrow A \rightarrow B \rightarrow t$ route, giving $x_1 = x_2 = 1$ and cost = $2 > \frac{3}{2}$.

Adding a road made everyone worse off. This is Braess's paradox (1968), and it has been observed empirically in Seoul, Stuttgart, and New York.

The **Price of Anarchy** quantifies the efficiency loss:

$$\text{PoA} = \frac{C(\text{Nash})}{C(\text{Social Optimum})}$$

For the Braess example: $\text{PoA} = 2/(3/2) = 4/3$.

i Why this matters

Braess's paradox shows that network structure interacts with strategic behaviour in non-obvious ways. Infrastructure investment, drug trafficking crackdowns (Dell 2015), and platform design all face versions of this problem. Understanding when adding capacity helps versus hurts requires understanding the network topology.

9 Summary and What's Next

9.1 What We Can Now Measure

Concept	Measure	Formula
Connectivity	Degree k_i	$\sum_j A_{ij}$
Average density	$\langle k \rangle$	$2m/n$
Global distance	$\langle \ell \rangle$, diam	BFS
Brokerage	Betweenness B_k	paths through k / all paths
Centrality	PageRank r_i	stationary random walk
Triangle closure	$C(A)$, $\bar{C}$	$\text{tr}(A^3)/\dots$

9.2 The Critical Gap: Where Do These Numbers Come From?

We can compute all of these measures on any real network. But then what?

Suppose we find that the American75 Facebook network has $C = 0.15$. Is that high? Low? Expected? To answer this, we need a **null model**; a probability distribution over graphs with the same basic parameters $(n, \langle k \rangle)$ against which to compare the observed statistic.

This is the core motivation for **random graph models**, the subject of Lecture 2.

Key questions that random graphs will let us answer:

1. What clustering coefficient should we expect from a network with n nodes and mean degree $\langle k \rangle$? (Erdős–Rényi model $\rightarrow$ answer: $C \approx 0$)
2. Why does clustering in real networks vastly exceed the ER prediction? (Watts–Strogatz small-world model)
3. Why do real networks have such heavy-tailed degree distributions? (Barabási–Albert preferential attachment)
4. Can we infer latent community structure from the observed edge pattern? (Stochastic Block Model)

```
n_fb <- vcount(g)
m_fb <- ecount(g)
p_er <- m_fb / (n_fb * (n_fb - 1) / 2)

set.seed(1)
g_er <- erdos.renyi.game(n = n_fb, p = p_er, type = "gnp", directed = FALSE)

# Helper: double-sweep on arbitrary graph
diam_g <- approx_diameter(g)
```



```

diam_g_er <- approx_diameter(g_er)

df <- data.frame(
  Metric = c("n", "m", "<k>", "Density", "C(A)", "C_bar", "Diameter (approx)"),
  American75 = c(
    vcount(g),
    ecountr(g),
    sprintf("%.2f", mean(degree(g))),
    sprintf("%.5f", graph.density(g)),
    sprintf("%.4f", transitivity(g, type = "global")),
    sprintf("%.4f", transitivity(g, type = "average")),
    diam_g
  ),
  ER_same_n_p = c(
    vcount(g_er),
    ecountr(g_er),
    sprintf("%.2f", mean(degree(g_er))),
    sprintf("%.5f", graph.density(g_er)),
    sprintf("%.4f", transitivity(g_er, type = "global")),
    sprintf("%.4f", transitivity(g_er, type = "average")),
    diam_g_er
  )
)

print(df, row.names = FALSE)

```

	Metric	American75	ER_same_n_p
	n	6386	6386
	m	217662	219007
	<k>	68.17	68.59
	Density	0.01068	0.01074
	C(A)	0.1635	0.0107
	C_bar	0.2474	0.0107
	Diameter (approx)	9	3

```
cat("\nConclusion: ER gets <k> right by construction, but clustering is ~10x too low.\n")
```

Conclusion: ER gets <k> right by construction, but clustering is ~10x too low.

```
cat("This gap is the central motivation for more realistic random graph models.\n")
```

This gap is the central motivation for more realistic random graph models.

9.3 References

- Easley, D., & Kleinberg, J. (2010). *Networks, Crowds, and Markets*. Cambridge University Press. Ch. 2 (Graphs), Ch. 3 (Triadic closure).
- Newman, M. E. J. (2013). *Networks: An Introduction*. Oxford University Press. Ch. 6 (Measures and metrics), Ch. 7 (Paths and distances), Ch. 8 (Clustering).
- Granovetter, M. S. (1973). The Strength of Weak Ties. *American Journal of Sociology*, 78(6), 1360–1380.
- Padgett, J. F., & Ansell, C. K. (1993). Robust Action and the Rise of the Medici. *American Journal of Sociology*, 98(6), 1259–1319.
- Rajkumar, K. et al. (2022). A causal test of the strength of weak ties. *Science*, 377(6612), 1304–1310.
- Braess, D. (1968). Über ein Paradoxon aus der Verkehrsplanung. *Unternehmensforschung*, 12(1), 258–268.